

Joshua Shinkle

jcshinkl@purdue.edu | TS/SCI Clearance

INTERESTS

I work in software/systems S&P. Specifically, I am interested in discovering vulnerabilities in mobile devices. My current research focuses on Android phones.

EDUCATION

- **Ph.D. in Computer Science advised by Antonio Bianchi**
 - Purdue University, *August '24 - May '29 (expected)*
- **B.S. in Computer Science - Cybersecurity**
 - Taylor University, *August '20 - May '24*

EXPERIENCE

Purdue University - Graduate Research Assistant - PurSec Lab *August '24 - Present*

- Developed a side-channel for Google Pixel's Edge TPU and the Apple Neural Engine (ANE)
- Conducted a user study investigating privacy risks in mobile apps marketed toward military-affiliated personnel
- Produced Android userspace exploits for DARPA's Intelligent Generation of Tools for Security (INGOTS) program
- Android app analysis
- Android OS reverse engineering

Peraton Labs - Cybersecurity Research Internship *Summer '25*

- Built an Android utility that leverages kernel tracing and interprocess communication introspection to trace messages between apps, telephony, Radio Interface Layer (RIL), and baseband firmware

Peraton Labs - Cybersecurity Research Internship *Summer '24*

- Trained a multitask deep learning recommender system to predict mobile app usage based on user demographics
- Worked as part of a team to repurpose mobile phone communication channels to enable secure communications

Peraton Labs - Cybersecurity Research Internship *Summer '23*

- Built a python tool to schedule/generate Android automations to mimic human activity
- Worked as part of a team to repurpose mobile phone communication channels to enable secure communications

Lockheed Martin ATL - Cybersecurity Research Internship *Summer '22*

- Reverse engineered Windows applications
- Windows kernel debugging